And then add another series as a line only, as shown below:

```
\addplot[line width=1pt]
coordinates {
    (0.1,0.0927143)
    (0.9,0.7937429)};
```

The final results are shown in Figures 2, 3 and 4.

Smoothing: an example of interacting with Scilab

If some calculations are necessary before plotting the data, Scilab (*http://www.scilab.org*) can be used. In the following script for Scilab, a moving average smoothing is calculated. The block length is equal to 2*m+1. A three-column text file called *smoothing.txt* is exported; the first column is an index from 1 to the data length, the second is the data values, and the third is the calculated moving average. The LaTeX engine is then called by the function *unix_g* and the script *smoothing.tex* produces the final plot (see Figure 5).

```
clear;
sheets=readxls(uigetfile("*.xls"));
data=sheets(1).value;
m=3;
[nr,nc]=size(data);
ma=zeros(1:nr);
for i=1+m:nr-m
    for j=-m:m
        ma(i)=ma(i)+data(i+j);
    end
    ma(i)=ma(i)/(2*m+1);
end
n=1:1:nr;
q=cat(2,n',data,ma');
write_csv(q,"smoothing.txt","    ",".");
unix_g("pdflatex smoothing.tex");
```

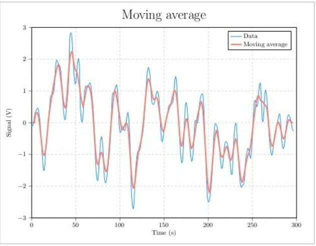

Figure 5: A plot from Scilab

Since in *smoothing.txt* the column headers are not present, in *smoothing.tex* each column is selected by its own index:

```
\addplot [p1] table[x index={0},y index={1}]{smoothing.txt};
\addplot [p2] table[x index={0},y index={2}]{smoothing.txt};
```

3D plot examples

The first thing I would like to talk about is how to create a custom colour map. Simply add the following code to *pgfplotsset* in the preamble:

```
colormap={parula}{rgb=(...,...,...)}
```

The complete RGB data for the *parula* and the *viridis* colour maps can be found in Reference 5. This technique is also useful to create a custom colour map for Scilab. In the following example, a 3D algebraic function is plotted and coloured with the *parula* colour map. For Scilab, also see Reference 6.

```
clear;
parula=[0.2081,0.1663,0.5292
        ...   ,...   ,...
        0.9763,0.9831,0.0538];
[nr,nc]=size(parula);
n=nr-1;
k=0:1:n;
cmap=interp1(k/n,parula,linspace(0,1,n));
deff("z=f(x,y)","z=sin(sqrt(x.^2+y.^2))");
x=-4:0.25:4;
y=-4:0.25:4;
xset("colourmap",cmap);
fplot3d1(x,y,f,alpha=80,theta=30);
s=gca();
s.box="off";
s.axes_visible="off";
xtitle("","","","");
```

The LaTeX code for the same function is a bit different, as seen below:

```
\documentclass[border=0.5cm]{standalone}
\usepackage{pgfplots}
\pgfplotsset{width=16cm,height=12cm,
    colourmap={parula}{
        rgb=(0.2081,0.1663,0.5292)
        rgb=(...    ,...    ,...    )
        rgb=(0.9763,0.9831,0.0538)
    }
}
\begin{document}
\begin{tikzpicture}
\begin{axis}[hide axis]
```